

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	1	1	1	4	2	2	1	1	1	1	1	1	3	3	2	2	2	1	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	4	2	4	4	1	3	3	3	3	3	3	1	1	4	4	2	4	4	2
Q.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	3	1	2	3	2	2	2	2	2	3	3	2	4	1	1	4	2	3	3	2
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80
A.	4	2	3	1	1	4	4	2	1	1	3	1	2	2	4	4	1	4	1	2
Q.	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
A.	3	1	3	4	1	4	4	1	1	1	4	1	2	3	4	1	1	2	2	4
Q.	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	3	1	3	2	1	3	2	2	3	2	2	2	3	3	1	3	4	4	2	2
Q.	121	122	123	124	125	126	127	128	129	130										
A.	4	1	3	1	2	2	3	2	2	3										

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HINT – SHEET

1. Ans (2)

Conceptual

2. Ans (1)

k_3 can be replaced by k'_3 at centre of rod, such that

$$k_3 \ell^2 = k'_3 \frac{\ell^2}{4}$$

also k_2 , k_3 are in series and their net in parallel with k_1

3. Ans (1)

Let the length of segment is " ℓ "

Let N is the no. of fringes in " ℓ "

and w is fringe width.

→ We can write

$$N w = \ell$$

$$N \left(\frac{\lambda D}{d} \right) = \ell$$

$$\frac{N_1 \lambda_1 D}{d} = \ell$$

$$\frac{N_2 \lambda_2 D}{d} = \ell$$

$$N_1 \lambda_1 = N_2 \lambda_2$$

$$16 \times 700 = N_2 \times 400$$

$$N_2 = 28$$

4. **Ans (1)**

The cell will deliver maximum power to the network if $R_{eq} = r$

$$\Rightarrow \frac{3R}{2} = r$$

$$\therefore R = \frac{2r}{3} = \frac{2}{3}\Omega$$

5. **Ans (4)**

At time



$$I = \frac{d}{dt} Q \Rightarrow V \frac{dc}{dt}$$

$$I = \frac{(k-1)Vbt_0}{d} = \text{const.}$$

6. **Ans (2)**

$$I = I_0 \sin t$$

$$(1) I_1 = \sqrt{\frac{\int_0^t I_0^2 \sin^2 t dt}{\int_0^t dt}} = \frac{I_0}{\sqrt{2}}$$

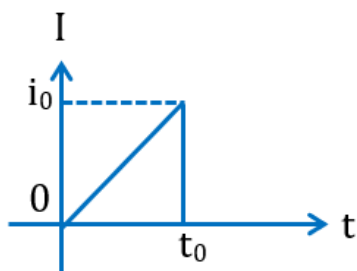
$$(2) I_2 = |I_0 \sin t|$$

$$I_2 = \sqrt{\frac{\int_0^t I_0^2 \sin^2 t dt}{\int_0^t dt}} = \frac{I_0}{\sqrt{2}}$$

$$(3) I_3 = \sqrt{\frac{\int_0^t I_0^2 dt}{\int_0^t dt}} = \frac{I_0 t}{t} = I_0$$

$$(4) I = \frac{I_0}{t_0} t$$

$$\begin{aligned} & \sqrt{\frac{\int_0^{t_0} I^2 dt}{\int_0^{t_0} dt}} = \sqrt{\frac{\int_0^{t_0} \left(\frac{I_0}{t_0} t\right)^2 dt}{t_0}} \\ & = \sqrt{\frac{\frac{I_0^2}{t_0^2} \frac{t_0^3}{3}}{t_0}} = \sqrt{\frac{I_0^2}{3}} = \frac{I_0}{\sqrt{3}} \end{aligned}$$



$$(B) I_3 > I_1 = I_2 > I_4$$

7. **Ans (2)**

For ring to be in equilibrium,

$$\mu mgr \geq Q \frac{\pi r^2 R}{2\pi r} \cdot r$$

8. **Ans (1)**

$$\text{Least count} = \frac{\text{Pitch}}{\text{total division on circular scale}}$$

In 1 revolution, it will travel 1 mm.

$$\text{So least count} = \frac{1}{50} = 0.02$$

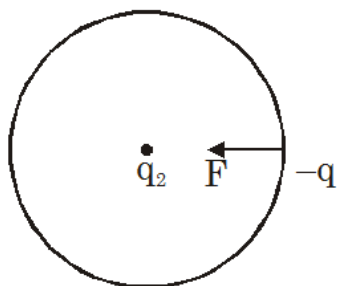
In 5 revolution, distance travel, 5 mm

9. **Ans (1)**

Field lines originate from +ve charge & terminates at -ve charge.

Also number of field lines are proportional to magnitude of charge.

10. **Ans (1)**



$$F = FC$$

$$\frac{kq_1q_2}{r^2} = \frac{mv^2}{r}$$

$$v = \sqrt{\frac{q_1q_2}{4\pi\epsilon_0rm}}$$

$$T = \frac{2\pi r}{v} = 2\pi r \sqrt{\frac{4\pi\epsilon_0m}{q_1q_2}}$$

$$T = \sqrt{\frac{16\pi^3m\epsilon_0r^3}{q_1q_2}}$$

11. **Ans (1)**

Wave is propagating in -ve x-direction.

$$\therefore \hat{E} \times \hat{B} = \hat{V}$$

$$\hat{K} \times \hat{B} = -\hat{i}$$

$$\text{So } \hat{B} = \hat{j}$$

$$B_0 = \frac{E_0}{C} = \frac{1.5 \times 10^6}{3 \times 10^8} = 5 \times 10^{-3} \text{ T}$$

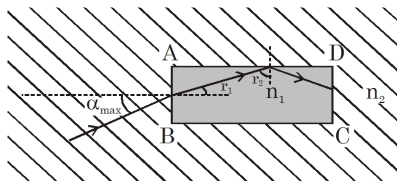
$$B = (B_0 \hat{j}) \sin[\omega t + kx + \phi]$$

Or

$$\vec{B} = (B_0 \hat{j}) \cos[\omega t + kx + \phi]$$

$$\vec{B} = (5 \times 10^{-3} \hat{j}) \cos(\omega t + kx)$$

12. Ans (1)



Total internally reflected at AD

$$\frac{n_1}{n_2} = \frac{\sin \alpha_{\max}}{\sin r_1} \Rightarrow \alpha_{\max} = \sin^{-1} \left[\frac{n_1}{n_2} \sin r_1 \right] \dots (i)$$

Also,

$$r_1 + r_2 = 90^\circ \Rightarrow r_1 = 90 - r_2 = 90 - C$$

$$\Rightarrow r_1 = 90 - \sin^{-1} \left(\frac{1}{2\mu_1} \right)$$

$$\Rightarrow r_1 = 90 - \sin^{-1} \left(\frac{n_2}{n_1} \right) \dots (ii)$$

Hence from equation (i) and (ii)

$$\begin{aligned} \alpha_{\max} &= \sin^{-1} \left[\frac{n_1}{n_2} \sin \left\{ 90 - \sin^{-1} \frac{n_2}{n_1} \right\} \right] \\ &= \sin^{-1} \left[\frac{n_1}{n_2} \cos \left(\sin^{-1} \frac{n_2}{n_1} \right) \right] \end{aligned}$$

13. Ans (1)

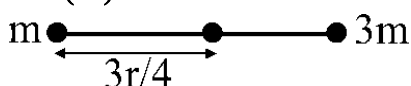
For retracing of light, light will fall vertically on the mirror.

By applying Snell's law $\mu_1 \sin(i) = \mu_2 \sin r$

$$(1) \sin 45^\circ = \mu \sin 30^\circ$$

$$\mu = \sqrt{2}$$

14. Ans (3)



$$\frac{Gm3m}{r^2} = m \left(\frac{3r}{4} \right) \omega^2 \Rightarrow \omega = \sqrt{\frac{4GM}{r^3}}$$

15. Ans (3)

The projectile is initially at semi-minor axis

$\therefore$ Satellite is at equal distance from both foci.

16. Ans (2)

friction force becomes zero, when normal contact force become zero. If v is the speed of particle at that moment, then

$$\frac{mv^2}{(R/2)} = qvB \Rightarrow v = \frac{qBR}{2m}$$

$\therefore$ change in K.E.

$$\Delta K = \frac{1}{2}m\left(\frac{qBR}{m}\right)^2 - \frac{1}{2}m\left(\frac{qBR}{2m}\right)^2$$

$$\Delta K = \frac{1}{2} \frac{q^2 B^2 R^2}{m} \left[1 - \frac{1}{4}\right]$$

$$\therefore \text{work done by friction} = -\frac{3}{8} \frac{q^2 B^2 R^2}{m}$$

$$\therefore K = 3$$

17. Ans (2)

$$\frac{hc}{\lambda_3} = \frac{hc}{\lambda_1} + \frac{hc}{\lambda_2}$$

$$\lambda_3 = \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$$

18. Ans (2)

The maximum K.E. of ejected photoelectron is $(K.E.)_{\max} = h\nu - \phi_0$

If the frequency of photon is doubled, maximum kinetic energy of photon electron becomes $(K.E.)'_{\max} = 2h\nu - \phi_0$

$$\frac{(K.E.)'_{\max}}{(K.E.)_{\max}} = \frac{2(h\nu - \frac{\phi_0}{2})}{h\nu - \phi_0} > 2$$

$$\text{Photo current} \propto \frac{\text{intensity of beam}}{h\nu}$$

If intensity and frequency both are doubled, the photocurrent remains same.

19. Ans (1)

Q value of the reaction

$$Q = (2 \times 4 \times 7.06 - 7 \times 5.6) = 17.28 \text{ MeV} \quad \dots(i)$$

$$K_P + Q = 2K_\alpha \quad \dots(ii)$$

$$\sqrt{2m_P K_P} = 2\sqrt{2m_\alpha K_\alpha} \cos \alpha$$

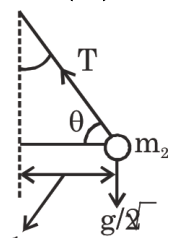
$$K_P = K_\alpha \quad \dots(iii)$$

$$\text{So } K_P = 17.28 \text{ MeV}$$

20. Ans (3)

$a_{\text{cm}} = 0$, v_{cm} may not be zero.

21. Ans (2)

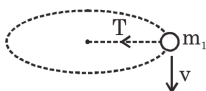


$$T \sin \theta = \frac{g}{\sqrt{2}}$$

$$T \cos \theta = \frac{1}{\sqrt{2}} \frac{(10)}{1} = \frac{10}{\sqrt{2}} = \frac{g}{\sqrt{2}}$$

$$\therefore \theta = 45^\circ$$

$$\Rightarrow T = g$$



$$T = \frac{0.1 \times v^2}{1} \Rightarrow v^2 = 100$$

$$\therefore v = 10 \text{ m/s}$$

22. Ans (4)

$$\text{Breaking stress} = \frac{F}{A}$$

If diameter is doubled, area becomes 4 times, hence force required will also become 4 times for same stress

23. Ans (2)

$$VT \propto r^2$$

24. Ans (4)

$$\Delta Q_{AB} = \Delta W_{AB} + \Delta U_{AB}$$

$$150 = 40 + \Delta U_{AB}$$

$$\Delta U_{AB} = 110 \text{ J}$$

$$\Delta U_{AB} + \Delta U_{BC} + \Delta U_{CA} = 0$$

$$110 + 50 + \Delta U_{CA} = 0$$

$$\Delta U_{CA} = -160$$

$$\Delta Q_{BC} = \Delta W_{BC} + \Delta U_{BC}$$

$$10 = \Delta W_{BC} + 50$$

$$\Delta W_{BC} = -40 \text{ J}$$

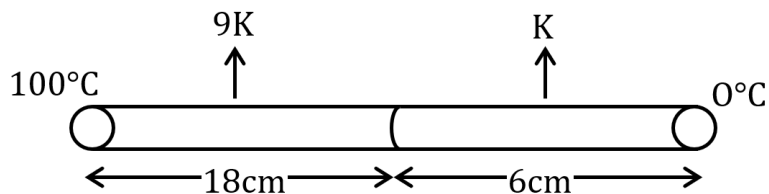
$$\Delta Q_{\text{net}} = \Delta W_{\text{net}} = 30 \text{ J}$$

$$\Delta Q_{CA} = -130 \text{ J}$$

25. Ans (4)

$$\beta = \frac{\Delta P}{\Delta V/V} = \frac{\delta gh}{0.001}$$

26. Ans (1)



$$\frac{Q}{t} = \frac{KA(T_1 - T_2)}{(T - 0)KA \frac{\ell}{6}} \Rightarrow \frac{(100 - T)9KA}{18}$$

$$300 - 3T = T$$

$$4T = 300$$

$$T = 75^\circ\text{C}$$

27. Ans (3)

Wien's displacement law $\lambda_m T = b$

28. Ans (3)

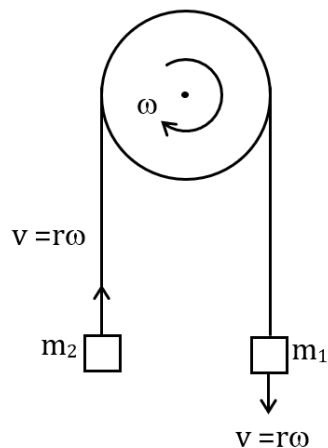
$$80 = \frac{1}{2} \times 10 \times t^2 \Rightarrow t = 4 \text{ sec}$$

$$10 \times 4 + v \times 4 = 60 \Rightarrow v = 5 \text{ m/s}$$

29. Ans (3)

since normal force is zero, thus friction force will also be zero.

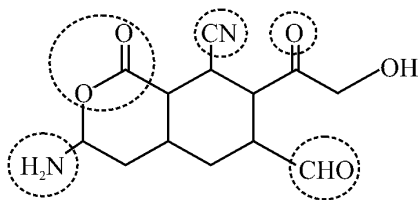
30. Ans (3)



$$L = m_1 vr + m_2 vr + I\omega$$

$$= (m_1 + m_2)r^2\omega + I\omega$$

31. Ans (3)



32. Ans (3)

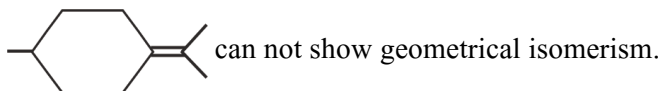
Stability of resonating structure : i > iii > iv > ii

(i) → Neutral

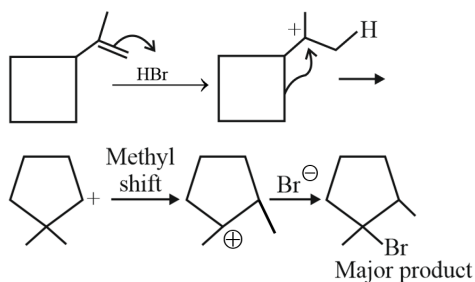
(iii) → Charged with complete octet

(ii), (iv) → Charged with incomplete octet

33. Ans (1)



34. Ans (1)



35. Ans (4)

We need inversion of configuration

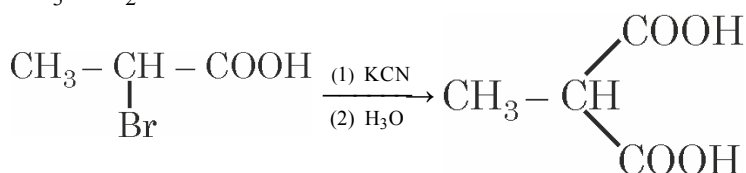
$\text{Br}_2 / \text{CCl}_4$ gives no reaction

SOBr_2 gives retention

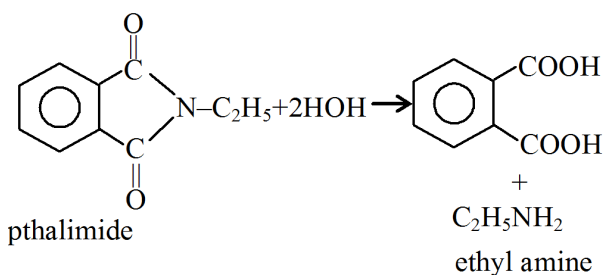
$\text{HBr} / \text{ZnBr}_2$ will give racemisation.

$\text{TsCl} / \text{Pyridine}$ followed by nucleophilic attack by Br^- will give inversion

36. Ans (4)

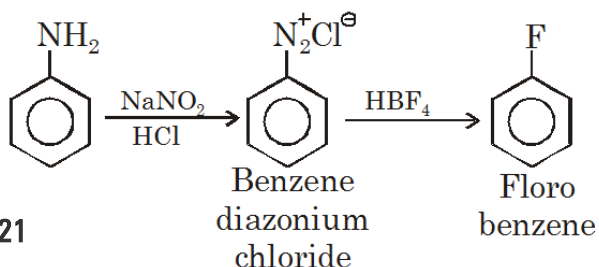


37. Ans (2)



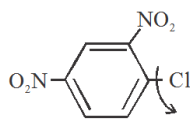
Therefore option (2) is correct.

38. Ans (4)



39. **Ans (4)**

Electron withdrawing group at ortho and para position increase rate of S_NAr



Easily replaced by $-OCH_3$ (nucleophing)

40. **Ans (2)**

Sucrose D(+) glucose + D(-) Fructose

$$[\alpha] = +66.5^\circ \quad [\alpha] = +52.5^\circ \quad [\alpha] = -92.8^\circ$$

Thus rotation changes from positive to negative after hydrolysis. Due to this reason hydrolysis of sucrose is known as inversion and mixture after hydrolysis is known as invert sugar.

Therefore option (B) is correct.

41. **Ans (3)**

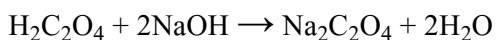
$$\text{Vol of Solution} = \frac{\text{mass}}{\text{density}}$$

$$= \frac{1000 + 120}{1.15} \text{ mL}$$

$$= \frac{1120}{1.15} \text{ ml}$$

$$\text{Molarity} = \frac{120/60}{\frac{1120}{1.15}} \times 1000 = 2.05 \text{ M}$$

42. **Ans (1)**



$$m_{eq} \text{ of } H_2C_2O_4 = m_{eq} \text{ NaOH}$$

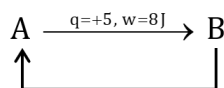
$$50 \times 0.5 \times 2 = 25 \times M_{NaOH} \times 1$$

$$\therefore M_{NaOH} = 2 \text{ M}$$

$$\text{Now } 1000 \text{ ml solution} = 2 \times 40 \text{ gram NaOH}$$

$$\therefore 50 \text{ ml solution} = 4 \text{ gram NaOH}$$

43. **Ans (2)**



$$\Delta U_{AB} = q + w = +5 + (-8) = -3$$

$$q = -3, \Delta U_{BA} = +3$$

$$\Delta U_{BA} = q + w$$

$$\Rightarrow 3 = -3 + w \Rightarrow w = +6 \text{ (work done on the system).}$$

44. **Ans (3)**

$$\Delta H = \Delta U + \Delta n_g RT \quad (\Delta n_g = -1)$$

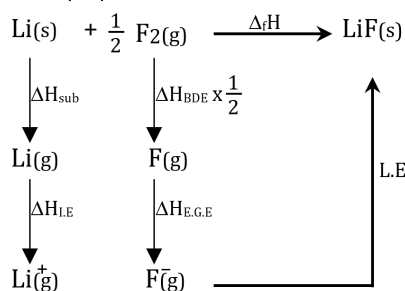
$$-1366.5 \text{ kJ/mole} = \Delta U - \frac{1 \times 8.314}{1000} \text{ kJ/mol} \times 300$$

$$-1366.5 = \Delta U - 2.5$$

$$-1366.5 + 2.5 = \Delta U$$

$$\Delta U = -1364.0$$

45. Ans (2)



$$\Delta_f H = \Delta_{\text{Hsub}} + \Delta_{\text{HIE}} + \Delta_{\text{H.BDE}} \times \frac{1}{2} + \Delta_{\text{H.E.G.E}} + \Delta_{\text{H.L.E}}$$

$$-617 = 161 + 520 + 77 + x - 1047$$

$$x = -328 \text{ kJ/mole}$$

46. Ans (2)

$$\begin{aligned}
 \frac{r_{\text{H}_2}}{r_{\text{O}_2}} &= \frac{n_{\text{H}_2}}{n_{\text{O}_2}} \sqrt{\frac{M_{\text{O}_2}}{M_{\text{H}_2}}} \\
 &= \frac{2}{1} \sqrt{\frac{32}{2}} = \frac{8}{1}
 \end{aligned}$$

47. Ans (2)

$$\text{rms} = \sqrt{\frac{u^2 + 16u^2 + 100u^2 + 225u^2}{4}}$$

$$\text{rms} = \sqrt{\frac{342}{4}} u$$

$$\text{Average} = \frac{u + 4u + 10u + 15u}{4} = \frac{30u}{4}$$

$$\text{ratio} = \frac{\frac{\sqrt{342}u}{2}}{\frac{30u}{4}} = \sqrt{\frac{342}{225}}$$

48. Ans (2)

$$r^+ + r^- = \sqrt{3} \frac{a}{2} = \frac{1.73 \times 4}{2} = 3.46 \text{ \AA}$$

49. Ans (2)

Non-spontaneous process at -10°C & 1 bar pressure.

50. Ans (3)

$$\begin{aligned}
 E_{\text{cell}} &= E_{\text{cell}}^0 - \log_{10} Q \\
 &= 0 - \frac{.06}{2} \log_{10} \left(\frac{[\text{H}_L^+]^2 P_{\text{H}_2\text{R}}}{P_{\text{H}_2\text{L}} [\text{H}_R^+]^2} \right) \\
 &= \frac{-0.06}{2} \log_{10} \left(\left(\frac{H_L^+}{H_R^+} \right)^2 \times \frac{2}{2} \right) \\
 &= -0.06 \log_{10} \left(\frac{10^{-5}}{10^{-7}} \right) = -0.12 \text{ volt}
 \end{aligned}$$

51. **Ans (3)**

$$\begin{aligned}\text{Number of moles of HCl added} &= \frac{0.73}{36.5} \\ &= 0.02 \text{ mol} = 20 \text{ m mol}\end{aligned}$$

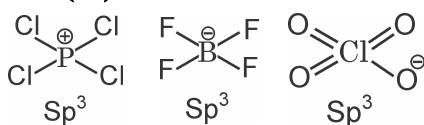
Then, number of millimoles required by

$$\begin{aligned}&1000 \text{ mL sol} \\ &= \frac{20}{200} \times 1000 = 100 \text{ m mol.}\end{aligned}$$

52. **Ans (2)**

$$\begin{aligned}\text{pH} &= 7 + \frac{1}{2} \{ \text{pK}_a + \log C \} \quad (\text{pK}_a(\text{HCN}) = 5.74) \\ &= 7 + \frac{1}{2} \{ 5.74 + \log(0.1) \} = 7 + \frac{1}{2} \{ 4.74 \} = 9.37\end{aligned}$$

53. **Ans (4)**

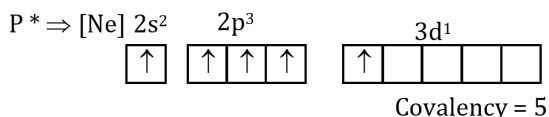
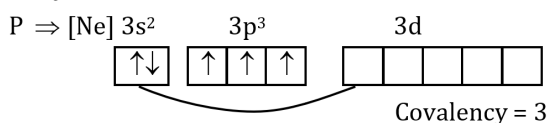


54. **Ans (1)**

PCl₅

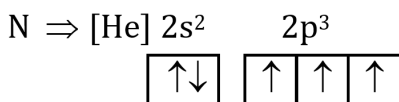
Due to presence of vacant orbital at low energy level, excitation is possible in P to form 5 bonds with 5 chlorine atoms

NCl₅



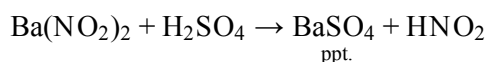
Due to absence of vacant orbital at low energy level excitation is not possible in N to form 5 bonds with 5 chlorine atoms.

NCl₅

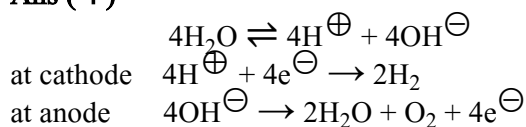


Due to absence of vacant orbital at low energy level excitation is not possible in N to form 5 bonds with 5 chlorine atoms.

55. **Ans (1)**

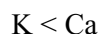


56. **Ans (4)**



By electrolysis of impure or acidified water, Very pure (99.9%) hydrogen (H₂ gas) can be obtained at cathode using inert solids like platinum/palladium as electrodes.

57. **Ans (2)**



V

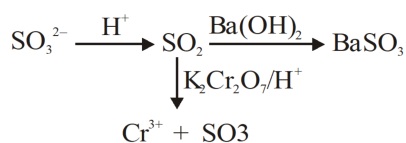
Ba

So ionisation potential order : **Ca** > Sr > Li > **Ba** > Na > **K**.

58. **Ans (3)**

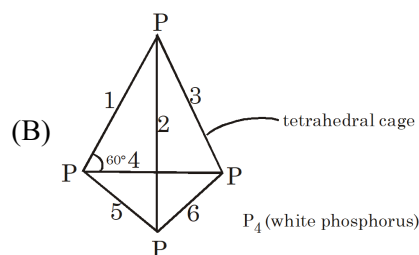
Melting point order is decided according to lattice energy but is also affected by polarisation, to higher L.E is of LiCl but also higher polarisation so, NaCl > KCl > CsCl > LiCl

59. **Ans (3)**



60. **Ans (2)**

(A) Black phosphorous is thermodynamically more stable allotrope of phosphorous



(C) Total 6 p-p bonds

(D) Total 12, 60° bond angles

So only option B is correct from given statements

71. **Ans (3)**

$$\text{As}, 7^3 + 7 = 350, 8^3 + 8 = 520$$

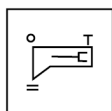
$$\text{As}, 5^3 + 5 = 130, 6^3 + 6 = 222$$

72. **Ans (1)**

First is made up of the second.

73. **Ans (2)**

By observation,



74. **Ans (2)**

By observation,



75. **Ans (4)**

By observation,



76. **Ans (4)**

Except option (4) in all other pairs first and second number multiplication value is third number.

77. **Ans (1)**

All have common alphabet R, E, A, T except TABLE.

78. **Ans (4)**

Only in fig. (4), the shadings occupy similar positions in the two halves of the circle.

79. **Ans (1)**

By observation,



80. **Ans (2)**



81. **Ans (3)**

$$1 + 1 = 2$$

$$1 + 2 = 3$$

$$2 + 3 = 5$$

$$3 + 5 = 8$$

$$5 + 8 = 13$$

$$8 + 13 = 21$$

$$Y - 1 = X$$

$$X - 3 = U$$

$$U - 8 = M$$

Hence, the correct answer is option (3).

82. **Ans (1)**

$$2^3 = 8 = H$$

$$2^4 = 16 = P$$

$$3^2 = 9 = I$$

$$1^4 = 1 = A$$

$$5^2 = 25 = Y$$

83. **Ans (3)**

$$(7^3+7), (6^3+6), (5^3+5), (4^3+4), (3^3+3), (2^3+2)$$

84. **Ans (4)**

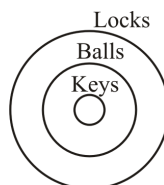
Total number of X

4, 6, 8, 10, 12

4 X should be there.

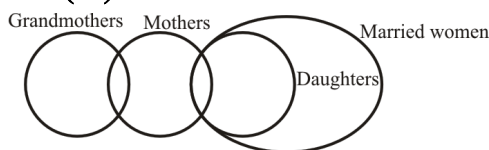
85. **Ans (1)**

From option (1), (A) and (C) are statement and (B) is conclusion, by venn diagram method we get



Hence conclusion, All keys are locks is true from above venn diagram. Hence, correct option is (1).

86. **Ans (4)**



Hence, only I and IV and either II or III follows.

87. **Ans (4)**

Statement of option (4) supports the passage.

88. **Ans (1)**

True progress in technology involves some risks too.

89. **Ans (1)**

By adding elements from column (1) & (2), we gets column (3).

90. **Ans (1)**

By observation,

Circle moving clockwise, arrow moving anti-clockwise and square moving clockwise in 3rd row to get the solution.

91. **Ans (4)**

$$\int \frac{1}{(\cos x)^{\frac{2}{3}} \cdot (\sin x)^{\frac{4}{3}}} dx$$

$$\int \frac{1}{\cos^2 x \cdot \left(\frac{\sin x}{\cos x}\right)^{\frac{4}{3}}} dx$$

$$\int (\tan x)^{-\frac{4}{3}} \cdot \sec^2 x dx$$

$$\tan x = t$$

$$\sec^2 x dx = dt$$

$$\int t^{-\frac{4}{3}} dt = \frac{t^{-\frac{1}{3}}}{-\frac{1}{3}} + c$$

$$= -3(\tan x)^{-\frac{1}{3}} + c$$

92. **Ans (1)**

$$\begin{aligned} & \lim_{x \rightarrow 3} \frac{x^3 - 7x^2 + 15x - 9}{x^4 - 5x^3 + 27x - 27} \\ &= \lim_{x \rightarrow 3} \frac{(x-3)(x^2 - 4x + 3)}{(x-3)(x^3 - 2x^2 - 6x + 9)} \\ &= \lim_{x \rightarrow 3} \frac{(x-1)(x-3)}{(x-3)(x^2 + x - 3)} \\ &= \lim_{x \rightarrow 3} \frac{x-1}{x^2 + x - 3} \\ &= \frac{3-1}{9+3-3} = \frac{2}{9} \end{aligned}$$

93. **Ans (2)**

Applying continuity at $x = -1$

$$-a + b = a + 1 + 3b$$

$$\boxed{2b + 2a + 1 = 0} \quad \dots(1)$$

Derivability at $x = -1$

$$a = 4a(-1) + 2(-1)$$

$$a = -4a - 2$$

$$5a = -2$$

$$\boxed{a = -\frac{2}{5}}$$

$$\text{put in (1) to get } b = \frac{-1}{10}$$

$$a + 2b = -\frac{4}{10} - \frac{2}{10} = -\frac{6}{10} = -\frac{3}{5}$$

94. **Ans (3)**

$$\frac{\sqrt{3x-7}}{(x+1)^{\frac{1}{4}} - 2}$$

for domain

$$3x - 7 \geq 0 \text{ \& } (x+1)^{\frac{1}{4}} - 2 \neq 0$$

$$x \geq \frac{7}{3} \text{ \& } x+1 \neq 16$$

$$x \geq \frac{7}{3} \text{ \& } x \neq 15$$

$$\left[\frac{7}{3}, 15 \right) \cup (15, \infty)$$

95. **Ans (4)**

$$\text{Let } f(x) = \sin(|x|) - |x|$$

$$\Rightarrow f(x) = \begin{cases} \sin x - x; & x \geq 0; \\ -\sin x + x; & x < 0 \end{cases}$$

$$\therefore f'(0^+) = 0 = f'(0^-)$$

$$\Rightarrow f(x) \text{ is differentiable at } x = 0.$$

96. **Ans (1)**

$$f(x) = x^3 + x^2 \quad x \in \left[-1, \frac{1}{2}\right]$$

$$f'(x) = 3x^2 + 2x = 0 \Rightarrow x = 0, \frac{-2}{3}$$

$$f(-1) = 0, f(0) = 0,$$

$$f\left(\frac{1}{2}\right) = \frac{3}{8}, f\left(\frac{-2}{3}\right) = \frac{4}{27}$$

97. **Ans (1)**

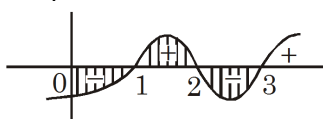
$$f(x) = xe^{x(1-x)}$$

$$f'(x) = e^{x(1-x)}(-2x-1)(x-1)$$

$$f'(x) \geq 0 \Rightarrow x \in \left(-\frac{1}{2}, 1\right)$$

98. **Ans (2)**

$$\begin{aligned} \text{Area} &= \left| \int_0^1 (x^3 - 6x^2 + 11x - 6) dx \right| \\ &+ \int_1^2 (x^3 - 6x^2 + 11x - 6) dx + \left| \int_2^3 (x^3 - 6x^2 + 11x - 6) dx \right| \\ &= \frac{11}{4} \end{aligned}$$



99. **Ans (2)**

$$\text{Let } x - [x] = t$$

$$x = t$$

$$\int_0^{\pi/6} \sec^2 x d(x - [x]) = \int_0^{\pi/6} \sec^2 t dt = \frac{1}{\sqrt{3}}$$

100. **Ans (4)**

(2, 3) belongs to R but (3, 2) is not

$\Rightarrow R$ is not symmetric

101. **Ans (3)**

Because inputs are less & outputs are more then no one-one onto function possible.

102. **Ans (1)**

$$y = (x-1)^3 - 1$$

$$x = (y+1)^3 - 1$$

$$f^{-1}(x) = 1 + \sqrt[3]{x+1}$$

103. Ans (3)

Let $x_1, x_2, \dots, x_n$ be n observations, then

$$\bar{X} = \frac{1}{n} \sum x_i$$

Let $y_i = \frac{x_i}{\alpha} + 10$, then

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n y_i = \frac{1}{\alpha} \left(\frac{1}{n} \sum x_i \right) + \frac{1}{n} (10n)$$

$$\Rightarrow \bar{Y} = \frac{1}{\alpha} \bar{X} + 10 = \frac{\bar{X} + 10\alpha}{\alpha}$$

104. Ans (2)

$$x^2 + bx + c = 0 \begin{matrix} \alpha \\ \beta \end{matrix}$$

$$\alpha + \beta = -b$$

$$\alpha\beta = c$$

$$\therefore (\sqrt{\alpha} + \sqrt{\beta})^2 = \alpha + \beta + 2\sqrt{\alpha\beta} = -b + 2\sqrt{c}$$

$$\Rightarrow \sqrt{\alpha} + \sqrt{\beta} = \sqrt{-b + 2\sqrt{c}}$$

105. Ans (1)

Using properties of A.P.,

$$a_1 \cdot a_{101} = 600$$

$$\text{we get } |a_1 - a_{101}| = 10.$$

106. Ans (3)

$$\frac{a(1 + r + r^2 + \dots + r^9)}{\frac{1}{a} + \frac{1}{ar} + \dots + \frac{1}{ar^9}} = a^2 r^9 = 3$$

Hence

$$(a)(ar)(ar^2) \dots (ar^9) = (a^2 r^9)^5$$

$$= (3)^5 = 243$$

107. Ans (2)

$$(b + \log_2 5)r = b + \log_4 5 \quad \dots (i)$$

$$(b + \log_4 5)r = b + \log_8 5 \quad \dots (ii)$$

$$(i) - (ii)$$

$$r \left[\log_2 5 - \frac{1}{2} \log_2 5 \right] = \frac{1}{2} \log_2 5 - \frac{1}{3} \log_2 5.$$

$$r \left[\log_2 \sqrt{5} \right] = \log_2 5 \left(\frac{1}{2} - \frac{1}{3} \right)$$

$$(5)^{\frac{r}{2}} = 5^{\frac{1}{6}} \quad r = \frac{1}{3}$$

108. Ans (2)

$$(x - 3)^2 = (x - 1)$$

$$\Rightarrow x^2 - 7x + 10 = 0$$

$$x = 2, 5$$

$x = 2$ is not possible

109. **Ans (3)**

Let the number is $x_1x_2x_3x_4x_5$.

Then $x_1 < x_2 < x_3 < x_4 < x_5$ with no digit can be zero.

$\therefore$ Total number of numbers $= {}^9C_5 = {}^9C_4$.

110. **Ans (2)**

$$(1+x)^{50} = 1 + {}^{50}C_1x + {}^{50}C_2x^2 + {}^{50}C_3x^3 + \dots + {}^{50}C_{49}x^{49} + {}^{50}C_{50}x^{50}$$

$\therefore$ Sum of coefficients of odd powers of x

$${}^{50}C_1 + {}^{50}C_3 + \dots + {}^{50}C_{49} = 2^{50-1} = 2^{49}$$

Hence (B) is the correct answer.

111. **Ans (2)**

We know that

$$\binom{n}{0}^2 + \binom{n}{1}^2 + \dots + \binom{n}{n}^2 = {}^{2n}C_n$$

so

$$({}^{12}C_0)^2 + ({}^{12}C_1)^2 + \dots + ({}^{12}C_{12})^2 = {}^{24}C_{12}$$

112. **Ans (2)**

$$A^T = A$$

$$(A^n)^T = (A^T)^n = (-1)^n A^n = -A^n; \quad \{n = \text{odd nature number}\}$$

113. **Ans (3)**

$$A - A^2 = I \Rightarrow A(I - A) = I \Rightarrow A - I = I - A$$

114. **Ans (3)**

$$\text{Use } R_2 \rightarrow R_2 - \left(\frac{R_1 + R_3}{2} \right)$$

115. **Ans (1)**

Here, $\Delta \neq 0$

$$\text{As, } \Delta = (23 - \alpha) \neq 0$$

116. **Ans (3)**

$$\sim(p \rightarrow q) = p \wedge \sim q$$

A quadrilateral is a square and it is not a rhombus.

117. **Ans (4)**

$$\frac{1 + \tan \theta}{1 - \tan \theta} + \frac{1 - \tan \theta}{1 + \tan \theta} = \lambda \left[\frac{1 + \tan^2 \theta}{1 - \tan^2 \theta} \right]$$

118. **Ans (4)**

$$\sin^3 x = \cos 2x$$

$$s^3 = 1 - 2s^2$$

$$s^3 + 2s^2 - 1 = 0$$

$$(s+1)(s^2 + s - 1) = 0$$

$$\sin x = -1 \text{ or } \sin x = \frac{-1 \pm \sqrt{5}}{2}$$

$$\text{only one solution in } \left(\frac{\pi}{2}, \pi \right) \text{ corresponding to } \sin x = \frac{\sqrt{5} - 1}{2}$$

119. Ans (2)

$$\sin^{-1}x + \sin^{-1}y = \frac{2\pi}{3}$$

$$\frac{\pi}{2} - \cos^{-1}x + \frac{\pi}{2} - \cos^{-1}y = \frac{2\pi}{3}$$

$$\cos^{-1}x + \cos^{-1}y = \frac{\pi}{3}$$

120. Ans (2)

Vector coplanar with given vectors

$$\vec{a} = x(2\hat{i} + \hat{j} + \hat{k}) + y(\hat{i} - \hat{j} + \hat{k})$$

$$\vec{a} = (2x + y)\hat{i} + (x - y)\hat{j} + (x + y)\hat{k}$$

$$\vec{a} \cdot (5\hat{i} + 2\hat{j} + 6\hat{k}) = 0$$

$$\Rightarrow y = -2x$$

$$\Rightarrow \vec{a} = x(2\hat{i} + \hat{j} + \hat{k}) - 2x(\hat{i} - \hat{j} + \hat{k})$$

Option (2) is correct

121. Ans (4)

$$\frac{-4+1}{5+4} = \frac{2-3}{5-2} = \frac{-2-2}{\lambda+2} \text{ or } \lambda+2=12 \text{ or } \lambda=10.$$

122. Ans (1)

$$c = -2a - 3b$$

$$ax + by - 2a - 3b = 0$$

$$a(x-2) + b(y-3) = 0$$

$$(x-2) + \lambda(y-3) = 0$$

$$x=2 \text{ \& } y=3$$

fixed point is (2, 3)

123. Ans (3)

Let point is R(x,y,z)

$$\therefore PR = QR$$

$$\Rightarrow (x-1)^2 + (y-2)^2 + (z-3)^2$$

$$= (x-3)^2 + (y-2)^2 + (z+1)^2$$

$$\Rightarrow x - 2z = 0$$

124. Ans (1)

equation of tangent to the parabola

$$y^2 = 4x \text{ is } y = mx + \frac{1}{m}$$

m is arbitrary constant

$$\frac{dy}{dx} = m$$

$$\Rightarrow y = x \frac{dy}{dx} + \frac{1}{\frac{dy}{dx}}$$

$$x \left(\frac{dy}{dx} \right)^2 - y \frac{dy}{dx} + 1 = 0$$

order = 1

125. Ans (2)

The given equation is $y = C_1 e^x + C_2 e^{-x}$

differentiate w.r.t. $x \Rightarrow \frac{dy}{dx} = C_1 e^x - C_2 e^{-x}$

Again differentiating $\Rightarrow \frac{d^2y}{dx^2} = C_1 e^x + C_2 e^{-x} = y$

$$\Rightarrow \frac{d^2y}{dx^2} - y = 0$$

126. Ans (2)

Given, $iz^3 + z^2 - z + i = 0$

Dividing both side by i and using $\frac{1}{i} = -i$, we have

$$z^3 - iz^2 + iz + 1 = 0$$

$$\Rightarrow z^2(z - i) + i(z - i) = 0 \quad (\because i^2 = -1)$$

$$\Rightarrow (z - i)(z^2 + i) = 0$$

$$\therefore z = i \text{ or } z^2 = -i$$

$$\therefore |z| = |i| = 1 \text{ or } z^2 = -i$$

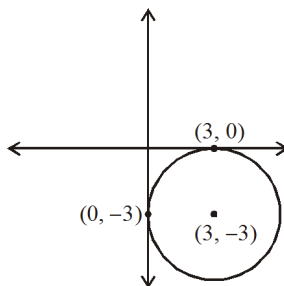
$$|z^2| = |-i| \Rightarrow |z|^2 = 1$$

$$\Rightarrow |z| = 1, -1$$

$$\therefore |z| = 1$$

127. Ans (3)

The complex numbers having least positive argument is $-3i$


128. Ans (2)

$P(WBWB) + P(BWBW)$

$$= \frac{5}{8} \times \frac{3}{8} \times \frac{5}{8} \times \frac{3}{8} + \frac{3}{8} \times \frac{5}{8} \times \frac{3}{8} \times \frac{5}{8} = 2 \left(\frac{15}{64} \right)^2 = \frac{15^2}{2^{11}}$$

129. Ans (2)

Total number of ways of taking out two socks is $^{12}C_2 = 66$.

The number of favourable ways

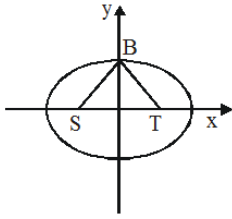
= Number of ways both the socks are
white or both are black

$$= {}^6C_2 + {}^6C_2 = 30.$$

$\therefore$ probability of required event

$$= \frac{30}{66} = \frac{5}{11}$$

130. Ans (3)



Let the equation of ellipse be $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

$$S \equiv (-ae, 0)$$

$$T \equiv (ae, 0)$$

$$B \equiv (0, b) \Rightarrow SB = \sqrt{(0 + ae)^2 + b^2}$$

$$\text{Also } SB^2 = ST^2 \Rightarrow 4a^2e^2 = a^2e^2 + b^2$$

$$\text{or } 3a^2e^2 = a^2(1 - e^2)$$

$$= a^2 - a^2e^2 \Rightarrow e^2 = 1/4 \Rightarrow e = 1/2$$

Hence (C) is the correct answer.